LESSON PLAN

AFCS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Homing and its forms
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Homing and its forms.
5. Specific Objectives	22.1.1.1Trainee will be able to understand the basic of guided weapon systems and types of targets.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of guided weapon system principles; Safety: No live equipment; Supervision: Instructor monitors understanding.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a homing guidance system?
(b) Name one type of target suitable for homing weapons.
(c) How does a homing weapon differ from a command-guided weapon?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Homing Guidance • Principle of Self-Steering • Basic Block Diagram Components • Active vs Passive Homing • Target Acquisition Mechanism	L & Ds	PPT, IB	15 min
(b)	• Forms of Homing: Active • Forms of Homing: Semi-Active • Forms of Homing: Passive • Advantages of Homing Systems • Limitations of Homing Systems	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Homing Types • Operational Scenarios • Target Types for Homing • Future Trends in Homing • Integration with Fire Control	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between active and passive homing in your own words.
(b) Identify two limitations of homing guidance systems discussed today.

12. Home Assignment and Follow Up	Prepare a brief note on the specific advantages and limitations of Infra-red homing to be discussed in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Infra-red homing, its block diagram and advantages and limitations of IR guidance Demonstration on Helmet mounted target designation system Film on Infra-red homing techniques	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min